Question 1

(a) False. Consider metric space $(\mathbb{R}, d_E)$ and define $A_i = [i, i+1]$. Since each A_i is closed and bounded, it is compact. But then $\bigcup_{n\in\mathbb{Z}} A_n = \mathbb{R}$ which is not bounded under $(\mathbb{R}, d_E)$ and thus not compact

(b) True. Take some $n \in I$. Then since A_n is compact, A_n is totally bounded. Let $\epsilon > 0$. Then there exists some finite set $\{x_1, x_2, ..., x_n\} \subseteq X$ such that $A_n \subseteq \bigcup_{i=1}^n B(x_i;\epsilon)$. But $\bigcap_{i\in I} A_i \subseteq A_n \subseteq \bigcup_{i=1}^n B(x_i;\epsilon)$. So $\bigcap_{i\in I} A_i$ is totally bounded. Now since each A_i is complete (follows from A_i being compact), $\bigcap_{i\in I} A_i$ is also complete. This follows from if (x_n) is Cauchy in $\bigcap_{i\in I} A_i$, then (x_n) is Cauchy in each A_i and thus converges to some $x \in A_i$ for all $i \in I$, so $x \in \bigcap_{i\in I} A_i$. So $\bigcap_{i\in I} A_i$ is totally bounded and complete, thus is compact

(c) False. Consider $f : ([1,\infty), d_E) \to (\mathbb{R}, d_E)$ defined by $f(x) = \frac{1}{x}$. First note that if $x_n \to x$, for $x_n, x \in [1,\infty)$. Then $|f(x_n) - f(x)| = |\frac{1}{x_n} - \frac{1}{x}| = |\frac{x_n - x}{x x_n}| \leq |x_n - x| \to 0$ as $n \to \infty$. So f is continuous. But then $[0,1]$ is compact under $(\mathbb{R}, d_E)$, and $f^{-1}([0,1]) = [1,\infty)$ which is not compact as it is not bounded

(d) False. Consider metric space $(\mathbb{R}, d_E)$ with $A_i = \{i\}$. Each A_i is connected. If $I = [-1,1] \cup [2,3]$, then $\bigcup_{i\in I} A_i = I$ which is disconnected so it is not connected

(e) False. Consider metric space $(\mathbb{R}^2, d_E)$. Let $A_1 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$ and $A_2 = \{(x,y) \in \mathbb{R}^2 : (x-1)^2 + (y-1)^2 = 1\}$. Now consider function $f : [0, 2\pi) \to \mathbb{R}^2$ defined by $f(\theta) = (2\cos\theta, 2\sin\theta)$. This is continuous because each component is continuous. This is a function on interval $[0, 2\pi)$ which is connected. Then $f([0,2\pi)) = A_1$ so A_1 is connected. Similarly, we can define $g : [0, 2\pi) \to \mathbb{R}^2$ by $g(\theta) = (1 + \cos\theta, 1 + \sin\theta)$ which is also continuous to get that $A_2 = g([0, 2\pi))$ is connected. But $A_1 \cap A_2$ is a set of 2 distinct points, say $A_1 \cap A_2 = \{(x_1, y_1), (x_2, y_2)\}$. This is then disconnected, as $\{(x_1, y_1)\}$ and $\{(x_2, y_2)\}$ are disjoint and closed and their union is $A_1 \cap A_2$, and thus not connected

Question 2

($\Rightarrow$) Suppose (X, d) is compact. Let (A_n) be a decreasing sequence of non-empty subsets of X. Now let x_n be some value in A_n for each $n \in \mathbb{N}$. Since (X, d) is compact, there exists some subsequence (x_{n_k}) such that $x_{n_k} \to x \in X$. Notice that since $x_n \in A_1$ for all $n \in \mathbb{N}$, we get that x_{n_k} is a sequence in A_1. Now let $N \in \mathbb{N}$. Now notice that sequence (x_{n_k+N}) is a subsequence of (x_{n_k}), and $n_k + N \geq k + N \geq N$, and since $x_N \in A_N$, and that $x_n \in A_N$ for all $n \geq N$, (x_{n_k+N}) is a sequence in A_N. But $x_{n_k+N} \to x$ as well (since it is a subsequence of (x_{n_k})), so $x \in \overline{A_N}$. Since this is true for every $N \in \mathbb{N}$, we get $x \in \bigcap_{n\in\mathbb{N}} \overline{A_n}$.
($\Leftarrow$) Suppose that for any decreasing sequence (A_n) of non-empty subsets of X, then $\bigcap_{n\in\mathbb{N}} \overline{A_n} \neq \emptyset$. Now let (x_n) be a sequence in X. Let $A_n = \{x_k : k \geq n\}$. This is a decreasing sequence of non-empty subsets of X. So there exists some $x \in \bigcap_{n\in\mathbb{N}} \overline{A_n}$. Then $x \in \overline{A_N}$ for each $N \in \mathbb{N}$. That means there exists some sequence $(y_{Nn})_{n=1}^{\infty}$ in A_N such that $y_{Nn} \to x$. We create subsequence (a_N) as follows: For each $N \in \mathbb{N}$, let $a_N = y_{Nn}$ for some y_{Nn} such that $d(y_{Nn}, x) < \frac{1}{N}$. Then (a_N) is a subsequence of (x_n), and $d(a_N, x) < \frac{1}{N}$ for all $N \in \mathbb{N}$, and since $\frac{1}{N} \to 0$ as $N \to \infty$, we get that (a_n) is a convergent subsequence of (x_n), converging to some $x \in X$. Thus (X, d) is compact